
Project Work Plan

Street View Robot

A mobile robot for 360° point-cloud reconstruction of indoor spaces

Project:	Street View Robot	Supervisor:	Mr. Rajaei Khatib
Team:	Group 9 (5 members)	Date:	31 May 2026
Platform:	Raspbot V2 / Raspberry Pi 5	Duration:	5 working days

1. Introduction

This document presents the work plan for the *Street View Robot* project. The objective of the project is to design and implement a mobile robot capable of autonomously traversing an indoor environment and constructing a three-dimensional point-cloud representation of its surroundings at each sampled location. Individual scans are subsequently merged into a single, room-scale 360° reconstruction.

The robot is built on the Raspbot V2 platform—a Raspberry Pi 5 chassis with four Mecanum wheels providing omnidirectional motion through an I²C motor driver. Depth sensing is provided by an Intel RealSense D405, a factory-calibrated stereo infrared camera. Additional onboard sensors (an ultrasonic rangefinder and four infrared line-tracking sensors) support obstacle detection and autonomous line following. The complete system is exposed through a Python hardware abstraction layer and supports two modes of navigation: manual keyboard control and autonomous line following.

2. Aim and Objectives

Aim. To deliver a functional mobile-robot system that incrementally builds a 360° point-cloud map of a room through coordinated navigation and stereo depth perception.

Objectives.

1. Assemble the robot chassis and configure the Raspberry Pi software environment.
2. Implement a hardware abstraction layer for movement, sensing, and image capture.
3. Compute depth from the stereo infrared pair and generate 3D point clouds.
4. Register and merge multiple views into a single 360° point cloud per location.
5. Implement the two navigation modes (manual control and autonomous line following).
6. Design and 3D-print a rigid mounting bracket for the depth camera.
7. Integrate, test, and document the complete system.

3. Scope and Deliverables

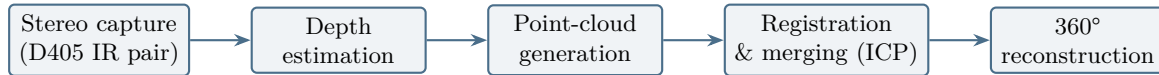
The project will deliver the following:

- An assembled, operational robot with the depth camera mounted on a custom 3D-printed bracket.
- A Python API abstracting all hardware into method calls (e.g. `forward()`, `capture_stereo()`, `set_tilt()`).
- A perception pipeline that produces a 360° point cloud at each sampled location and merges locations into a room-scale scan.

- Two functional navigation modes: manual (keyboard) and autonomous (line following).
- Source code, technical documentation, and a final demonstration.

4. Methodology

The technical approach is organised around the data-processing pipeline shown below.



4.1 Depth Estimation

Depth is computed from the rectified left/right infrared images using block-matching (semi-global block matching). Disparity is converted to metric depth using the camera's factory-calibrated focal length and stereo baseline ($Z = f_x \cdot b/d$). The result is validated against the camera's built-in depth stream.

4.2 Point-Cloud Generation

Each depth image is back-projected into 3D space using the pinhole camera model and the factory intrinsics, producing one point cloud per viewpoint.

4.3 Registration and Merging

At each location the robot rotates through a sequence of known angular steps, capturing a point cloud at each step. Clouds are aligned using the Iterative Closest Point (ICP) algorithm, initialised with the robot's known rotation, and combined into a single 360° point cloud. Outlier removal and voxel down-sampling are applied to control noise and density.

4.4 Navigation and Control

Omnidirectional motion is achieved through Mecanum-wheel kinematics issued over I²C. In manual mode the robot is driven via keyboard input; in autonomous mode it follows a taped path using its four infrared sensors and halts at cross-shaped stop markers to trigger the capture routine.

4.5 Camera Mounting

A rigid bracket will be designed and 3D-printed to fix the D405 (42 × 42 × 23 mm, 58 g, secured by two M3 screws) to the chassis at a fixed, level, and repeatable pose—a prerequisite for accurate point-cloud registration.

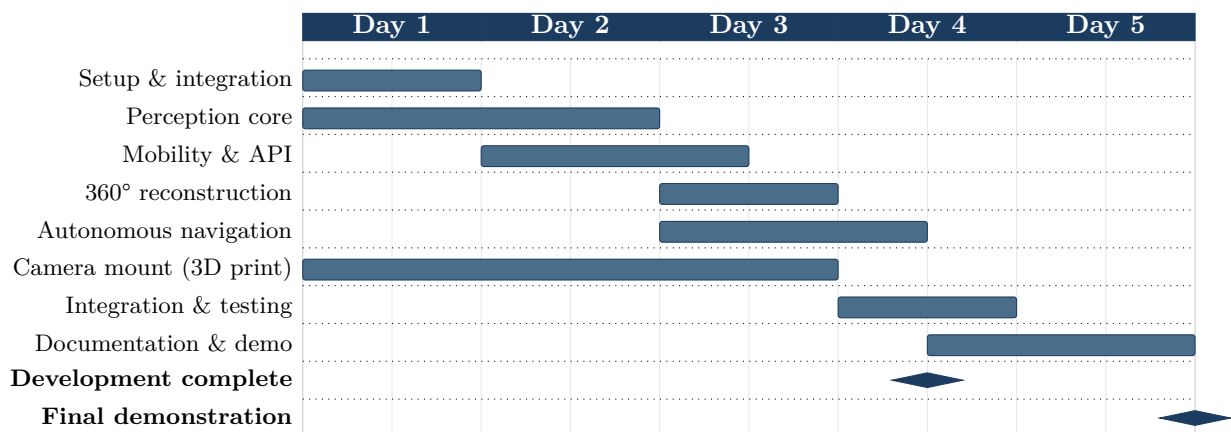
5. Work Plan and Schedule

The project is scheduled across five working days. Development and implementation are to be completed by the **middle of Day 4**; the remaining time is allocated to integration, testing, and preparation of the final demonstration.

5.1 Work Breakdown

ID	Phase	Tasks	Duration
1	Setup & Integration	Chassis assembly; Raspberry Pi OS, I ² C and camera configuration; software environment; network access.	Day 1
2	Perception Core	Stereo depth estimation; point-cloud generation; validation against factory depth.	Day 1–2
3	Mobility & API	Hardware API; Mecanum motion; servo and sensor control; manual navigation mode.	Day 2–3
4	360° Reconstruction	Rotate–capture–merge routine; ICP registration; single-location 360° cloud.	Day 3
5	Autonomous Navigation	Line-following control; stop-marker detection; capture-routine integration.	Day 3–4
6	Mechanical	Camera-mount measurement, design, 3D printing and fitting.	Day 1–3
7	Integration & Testing	Multi-location collection; system testing; refinement.	Day 4
8	Documentation & Demo	Final documentation; demonstration preparation.	Day 4–5

5.2 Schedule (Gantt Chart)



Each day comprises two half-day units. Development is completed at the middle of Day 4 (“Development complete” milestone).

6. Milestones

No.	Milestone	Target
M1	Robot assembled, Pi configured, camera capturing frames	End of Day 1
M2	Depth estimation and point-cloud generation operational; manual driving working	End of Day 2
M3	360° point cloud produced at a single location; camera mount fitted	End of Day 3
M4	Both navigation modes integrated; development complete	Mid Day 4
M5	System tested, documented, and demonstrated	End of Day 5

7. Resources and Tools

- **Hardware:** Raspbot V2 chassis, Raspberry Pi 5, Intel RealSense D405, ultrasonic and infrared sensors, 3D printer.
- **Software:** Python 3, OpenCV, NumPy, pyrealsense2, Open3D; Raspberry Pi OS (64-bit); CAD/slicing software for the mount.

8. Risk Assessment

Risk	Impact	Mitigation
RealSense driver build fails on Raspberry Pi	High	Allow buffer time on Day 1; develop perception on a laptop in parallel.
Poor depth on low-texture surfaces (passive stereo)	Medium	Use textured, well-lit capture environments; tune matching parameters.
Inaccurate cloud registration / merging	Medium	Use small rotation steps with high overlap; initialise ICP with the known rotation.
3D-print failure or delay	Medium	Begin design on Day 1; prepare a temporary mounting fall-back.
Network access to the robot unavailable	Low	Use an alternative local network (mobile hotspot) for SSH access.
Schedule overrun	Medium	Prioritise manual mode (essential); treat autonomous mode as secondary; enforce the Day-4 development freeze.

9. Expected Outcomes

On completion, the project is expected to deliver a fully operational robot that navigates an indoor space, autonomously captures and reconstructs 360° point clouds at multiple locations, and merges them into a coherent room-scale map. Both navigation modes will be demonstrated, and the depth camera will be mounted on a purpose-built 3D-printed bracket. All source code and documentation will be delivered, accompanied by a final demonstration to the supervisor.